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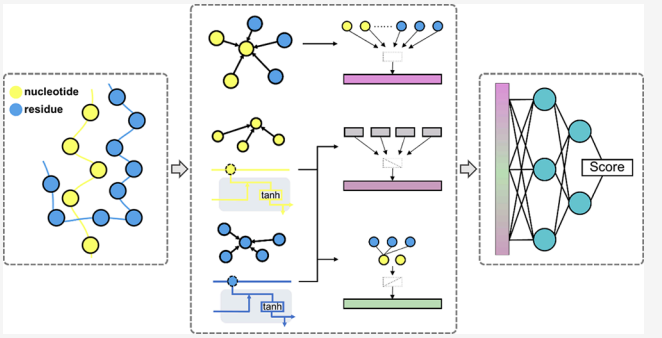
Graph Learning-Based Scoring of RNA−Protein Complex Structures

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ABSTRACT: Development of suitable scoring functions is essential for the prediction of RNA−protein complex structures. Conventional statistical potential-based scoring functions suffered from deficiencies in handling conformational flexibility. The recent application of convolutional neural network (CNN) to this field has shown the potential to address the problem. Compared to CNN, however, the graph deep learning generally exhibited better performance for biomolecule-related structural and functional prediction tasks. Herein, we propose EGARPS+, a novel attempt to apply the graph learning theory to evaluate RNA−protein complex structures. This algorithm comprised the intermolecular and intramolecular modules, which were established on the equivariant graph neural networks and specifically designed attention mechanisms. Additionally, we adopted previously unexplored sequence, structural and interaction features to fully represent interface regions. Our algorithm consistently outperformed the CNN-based approach and traditional statistical potentials on both bound and unbound data sets. The proposed model excelled in processing complexes with larger conformational changes, smaller interface sizes, and lower structural similarities. EGARPS+ could also improve the de novo RNA−protein complex prediction by RoseTTAFoldNA and AlphaFold3. Finally, interpretability analyses underscored the importance of conserved motifs and hydrogen bonding in RNA−protein interactions.



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INTRODUCTION

The interaction between RNA and protein plays a key role in biological processes, such as post-transcriptional regulation and protein synthesis.[1−4] The dysregulation of these interactions could result in different diseases, including human genetic disorders and cancer.[5−7] Determination of RNA−protein complex structures is helpful in uncovering the molecular mechanisms of these interactions.[8] However, the number of known RNA−protein complex structures is very limited, probably because the experimental techniques (e.g., X-ray crystallography, NMR spectroscopy and cryo-electron microscopy) are time-consuming and costly.[9−11] The emerging computational methods may be an alternative way to solve this problem. Especially, molecular docking is the most representative method, which could be divided into two steps, namely the sampling of possible conformations and the scoring of generated conformations.[12,13] Therefore, development of reliable scoring functions is essential for accurately identifying native-like conformations.[14,15]

Several scoring functions have been developed during the past decade and could be classified into two categories, namely statistical potential- and machine learning-based functions. Regarding the first category, the preference of interfacial contacts in known complexes was analyzed at the atomic or residue level, and then the potential function could be constructed by considering different knowledge-based terms. For example, Perez-Cano et al. proposed a propensity-based function which integrated the propensity of interfacial residue-nucleotide pairs with the inverse Boltzmann formula.[16] Tuszynska et al. developed two medium-resolution potentials, named DARS-RNP and QUASI-RNP, through decomposing nucleotides and residues into united atoms and designing four energy terms in terms of distances, angles, interaction sites, and steric clashes.[17] Li et al. designed a scoring function called 3dRPC-Score that treated the conformations of residue-nucleotide pairs as statistical variables and considered both distance and orientation between residues and nucleotides.[18] Huang and Zou established a potential function termed ITScore-PR which used an iterative strategy to update the distance-dependent potentials by comparing the distribution functions of predicted atomic pairs with those of native pairs.[19] Generally, the atomic-level potentials could exhibit better discriminative abilities, while the residue-level potentials might be more tolerant to the conformational changes upon binding.

Recently, machine learning techniques were also applied to the design of scoring functions. Parisien et al. proposed a chemical context profile-based scoring function that exploited

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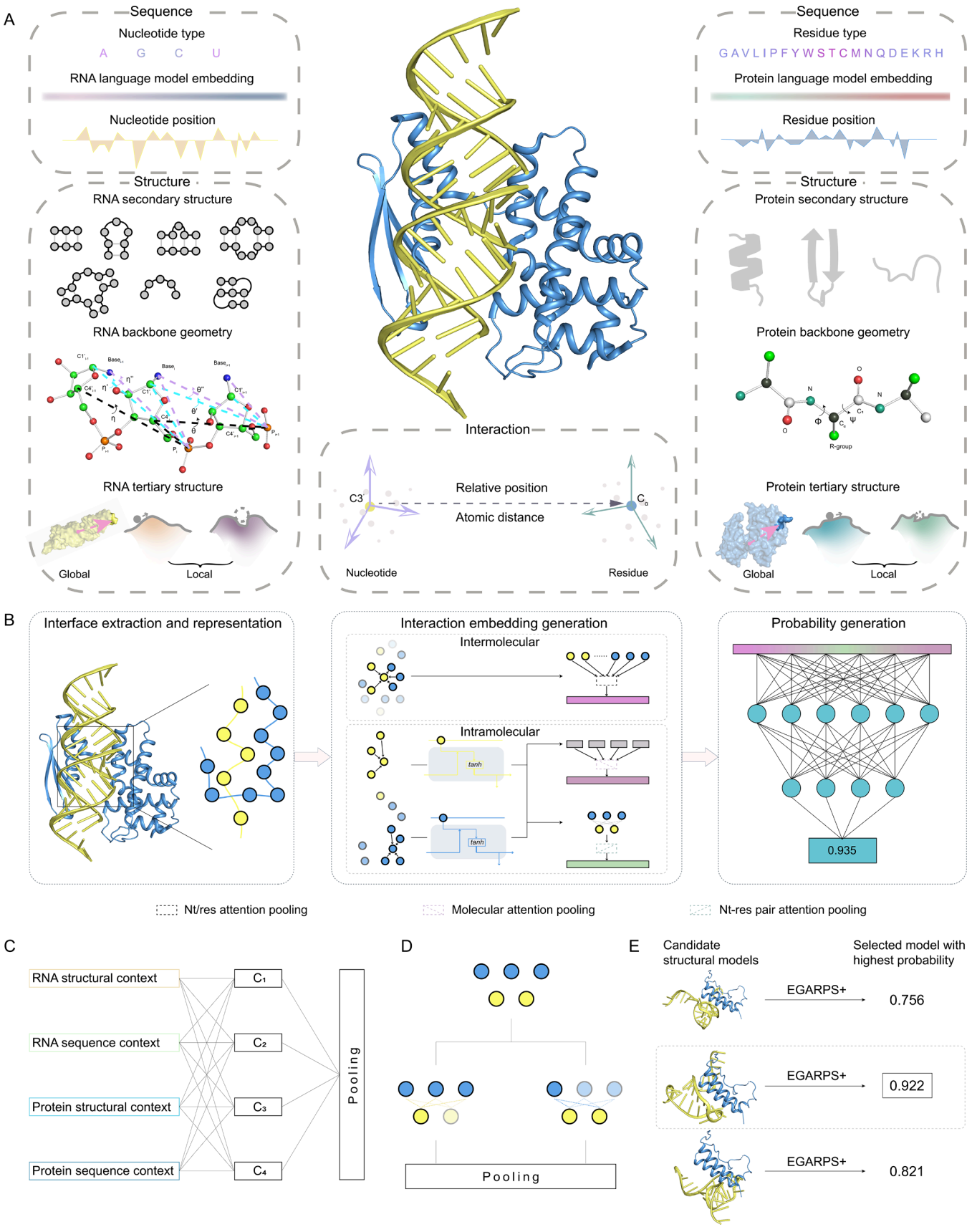


Figure 1. Features used to represent an interface and the flowchart of proposed algorithms. (A) Input features include sequence and structure characteristics of RNA and protein molecules, as well as interaction features between these two molecules. EGARPS uses relative geometry properties as interaction features, while EGARPS+ additionally adopts atom-based distances. (B) Our algorithm takes interface representations as input, generates interaction embedding by intermolecular and intramolecular modules, and outputs a prediction probability using a fully connected

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Figure 1. continued

layer. This process is repeated once and three times for EGARPS and EGARPS+, respectively. The average of the outputs of three models with different seeds is the final result of EGARPS+. (C) Schematic of molecular attention pooling. (D) Schematic of nt-res pair attention pooling. (E) Our algorithm evaluates each candidate structural model and selects the model with the highest probability.

machine learning methods to extract critical pairs of moieties involved in RNA−protein binding and determine the weights of elements in the profile.[20] Zeng et al. developed the first deep learning-based scoring function, termed DRPScore, which used a 4D convolutional neural network (CNN) to learn the binding patterns of interfacial atoms.[21] Moreover, they claimed that their method outperformed statistical potential-based approaches. Despite the contribution of DRPScore, there is still room for improvement. First, graph deep learning techniques (e.g., graph neural networks) have been widely applied to biomolecule-related structural and functional prediction problems and generally exhibited better performance than CNN-based models.[22−26] For instance, several recent models, such as EquiDock and RNAformer, have adopted equivariant networks or transformers in RNA and RNA−protein structural modeling.[27,28] However, graph-based methods have yet to be developed for evaluating RNA−protein complexes. Second, DRPScore only used the occupation number, mass, and charge of interfacial atoms as input features and therefore lacked an in-depth characterization of binding regions. In our previous works, we designed a series of biophysically inspired descriptors to predict the binding sites in proteins and RNAs.[29−32] These features could be extended to the scoring of RNA−protein interfaces. Moreover, incorporation of intermolecular interaction features may also strengthen the utility of scoring functions.

Motivated by the above issues, we present EGARPS+, an equivariant graph neural network and attention mechanism-based RNA−protein complex scoring function. To effectively capture the binding patterns, we designed an intramolecular module and an intermolecular module. Meanwhile, we adopted both sequence and structural properties of individual molecules as well as interaction properties between molecule pairs to represent interfaces. Combining graph learning techniques with these designs provided more accurate evaluations of RNA−protein complex structures. Extensive comparisons illustrated that EGARPS+ achieved consistently superior performance compared with state-of-the-art methods, especially on unbound data sets. Additionally, the top 1 success rate of our algorithm was more than twice that of the CNN-based method. Our graph-based scoring function could improve the performance of de novo prediction of RNA−protein complexes. The interpretability of our algorithm suggested the importance of conserved motifs and hydrogen bonding in RNA−protein interactions.

RESULTS

Overview of the EGARPS Series. Figure 1 and Supplementary Figure 1 show the framework of EGARPS series. We comprehensively extracted intramolecular features and intermolecular interaction features to depict RNA−protein binding patterns (Figure 1A). The intramolecular features referred to the descriptors derived from the primary, secondary and tertiary structures of each binding partner, while the intermolecular interaction features included relative geometric properties and atom-based distances (the latter was only adopted by the enhanced version, EGARPS+). Our algorithm took the interface of an RNA−protein complex as input and yielded a prediction probability of being a native-like complex (Figure 1B). Here, an interface included a set of nucleotide-residue pairs within a distance of 4.5 Å. Then, two different approaches, namely the intermolecular and intramolecular modules, were designed to generate the embedding of an interface. In the intermolecular module, the equivariant graph neural network (EGNN) and nt/res attention pooling were applied to the graph representation of interfacial nucleotides and residues.[33,34] In the intramolecular module, the EGNN and gate recurrent unit (GRU) were separately adopted to update the residue/nucleotide representations.[35] The former could aggregate the information on spatially neighboring positions, while the latter could capture the latent sequential patterns. The outputs from EGNNs and GRUs were processed by two attention pooling layers. For the molecular attention pooling, structural and sequence contexts were generated to represent the interface, and attention operation was performed between these contexts to capture the shape complementarity and pattern matching between binding partners (Figure 1C). For the pair attention pooling, attention was performed on all nucleotides and residues to find the crucial pairs that support RNA−protein interactions (Figure 1D). Finally, the resultant embedding of an interface was fed into a fully connected layer to estimate the prediction score. For EGARPS+, we repeated the above procedures three times with different initial seeds and took the average score as the final result. More details are presented in the Materials and Methods section and Supplementary Figure 2.

Performance of Different Methods on Testing Sets. We evaluated EGARPS and EGARPS+ on three testing sets and compared them with five existing methods, including one deep learning-based method (DRPScore), two atom-based statistical potential methods (ITScore-PR and DITScorePR) and two residue-based statistical potential methods (3dRPC and DARS-RNP).[17−19,21,36] Figure 2A−C shows the success rates of different methods from the top 1 to top 1000 predictions. On the bound data set (testing set I), the EGARPS series and atom-based statistical potential methods demonstrated better performance than DRPScore and residue-based statistical potential methods. Specifically, EGARPS+ achieved the highest top 1 success rate of 73.5%, followed by DITScorePR with a success rate of 69.4%, while DRPScore performed less well with a value of 32.6%. On the unbound data set (testing set II), the similar tendency was observed for these methods. Top 1 success rates for EGARPS+ and EGARPS were 35.09% and 31.58%, respectively. Although DITScorePR shared the same top 1 success rate with EGARPS +, our approaches clearly outperformed other methods from the top 30 to top 300 predictions. Furthermore, EGARPS series showed remarkable advantages over competing methods on the more rigorous unbound data set (testing set III). The performance difference between atom- and residue-based statistical potential methods was reduced for these samples, while DRPScore, the CNN-based approach, achieved the worst performance, with a top 1 success rate of less than 5%. Furthermore, by comparing the complexes having a successful

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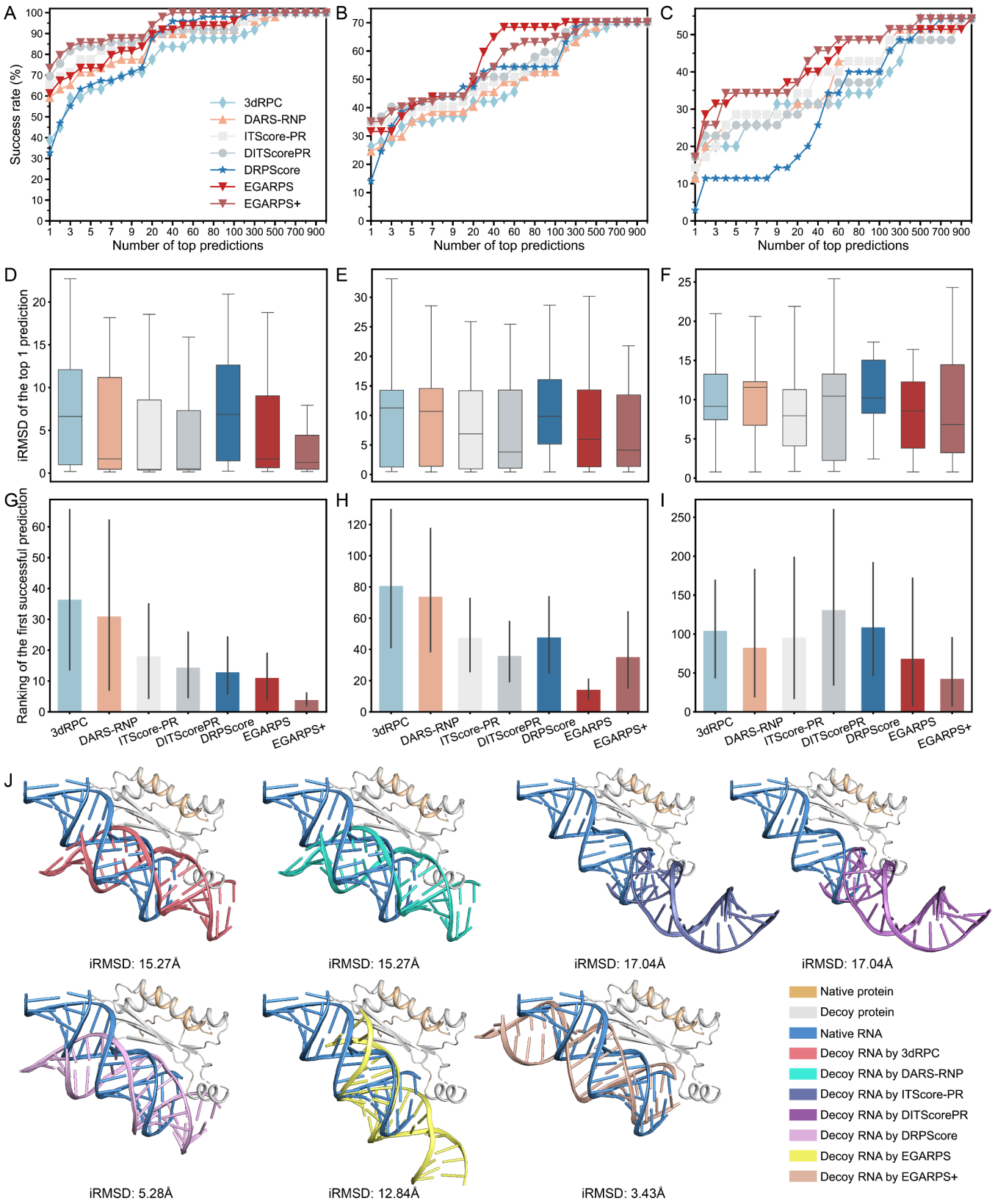


Figure 2. Performance comparison between EGARPS series and other methods on testing sets. (A) Success rates on testing set I. (B) Success rates on testing set II. (C) Success rates on testing set III. (D) Distribution of iRMSD for top 1 predictions on testing set I. (E) Distribution of iRMSD for top 1 predictions on testing set II. (F) Distribution of iRMSD for top 1 predictions on testing set III. (G) Average ranking of the first successful predictions on testing set I. (H) Average ranking of the first successful predictions on testing set II. (I) Average ranking of the first successful predictions on testing set III. (J) Top 1 predictions of an example (PDB ID: 1R9F) generated by different methods.

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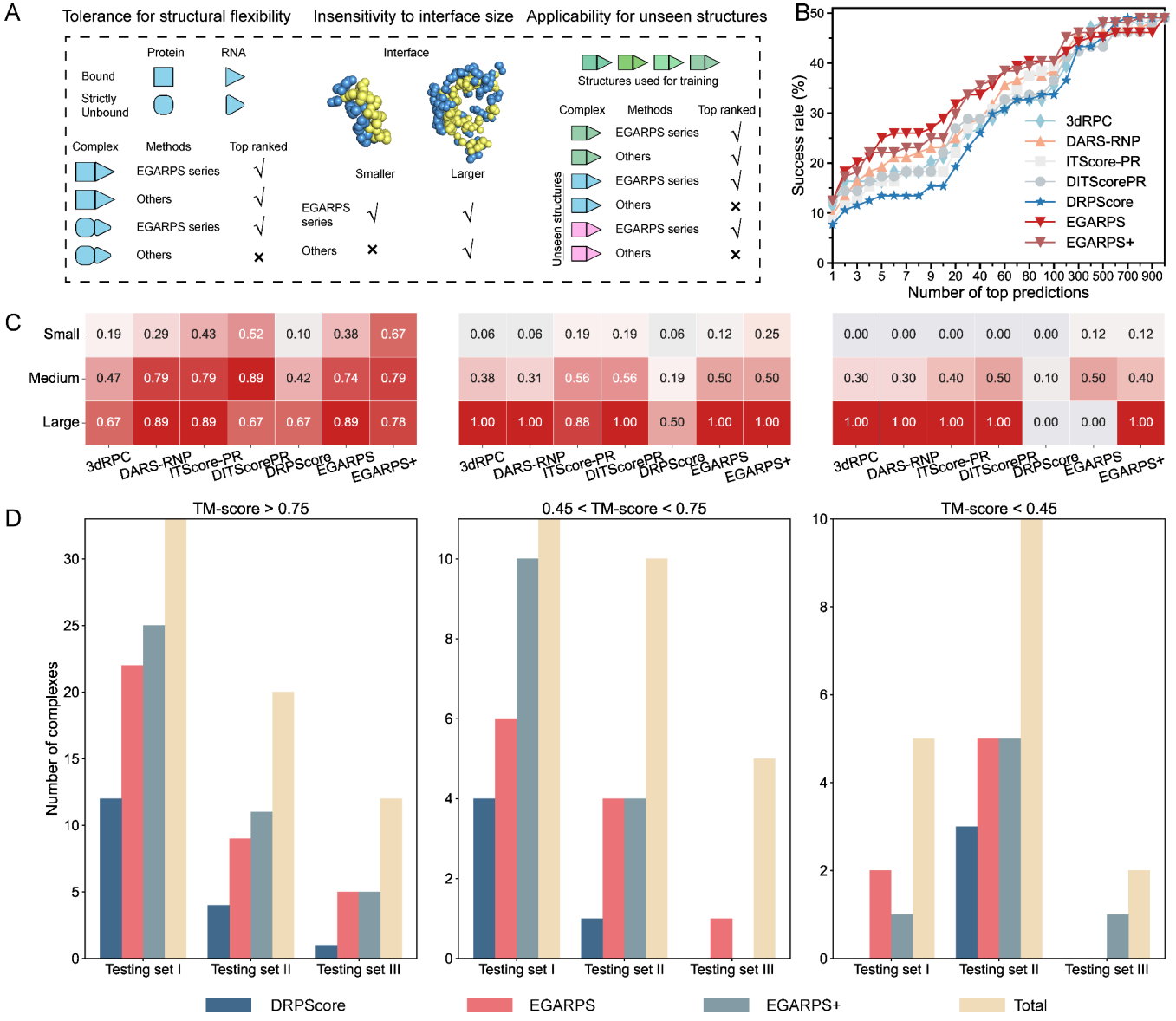


Figure 3. Advantages of our method over existing algorithms. (A) A summary of the advantages of EGARPS series from three perspectives. (B) Success rates on the docking benchmark built by Perez-Cano et al.'s work. (C) Success rates of top 1 prediction for complexes with different interface sizes in testing sets I, II, and III, respectively. (D) Numbers of successfully predicted complexes in three testing sets based on different structural similarities.

top 1 prediction, we found that our methods provided a complement to existing methods (Supplementary Figure 3). As shown in Figure 2D−F, the EGARPS series achieved relatively lower iRMSD measures for the top 1 prediction compared to other methods. The median of EGARPS+ is less than 5 Å on testing sets I and II and slightly greater than 5 Å on testing set III. Figure 2G−I displays the rankings of the first successful prediction for each method. EGARPS series yielded the lowest average rankings, demonstrating its superior predictive capability. On testing set III, especially, EGARPS+ was the only algorithm achieving an average ranking within the top 50. The iRMSD and ranking of the first successful prediction of each complex in the three testing sets are provided in Supplementary Tables 1−3.

Figure 2J shows the top1 predictions of a complex selected by different methods. EGARPS+ generated the lowest iRMSD of 3.43 Å, compared to 5.28 Å for DRPScore and greater than 10 Å for the remaining algorithms. Another example is presented in Supplementary Figure 4. In this case, EGARPS and EGARPS+ yielded an identical iRMSD of 2.02 Å, whereas the iRMSD of all other methods exceeded 15 Å. Notably, in testing sets II and III, there were 17 and 16 complexes, respectively, whose iRMSD values of all decoys were greater than 4 Å. Despite the higher deviations, certain decoys may be physically reasonable, because a considerable proportion of native binding sites could be retrieved for the complex structure (Supplementary Figure 5A). Therefore, we analyzed the best ranking of decoys with an iRMSD ranging from 4 Å to 8 Å. As shown in Supplementary Figure 5B,C, the EGARPS series achieved the best results among all methods on testing set II and outperformed 3dRPC and DARS-RNP on testing set III. These analyses demonstrate the ability of EGARPS series to effectively identify the native-like complexes that are composed of (rigorous) unbound structures.

Advantages of Our Algorithm over Other Methods. As shown in Figure 3A, the advantage of EGARPS series over

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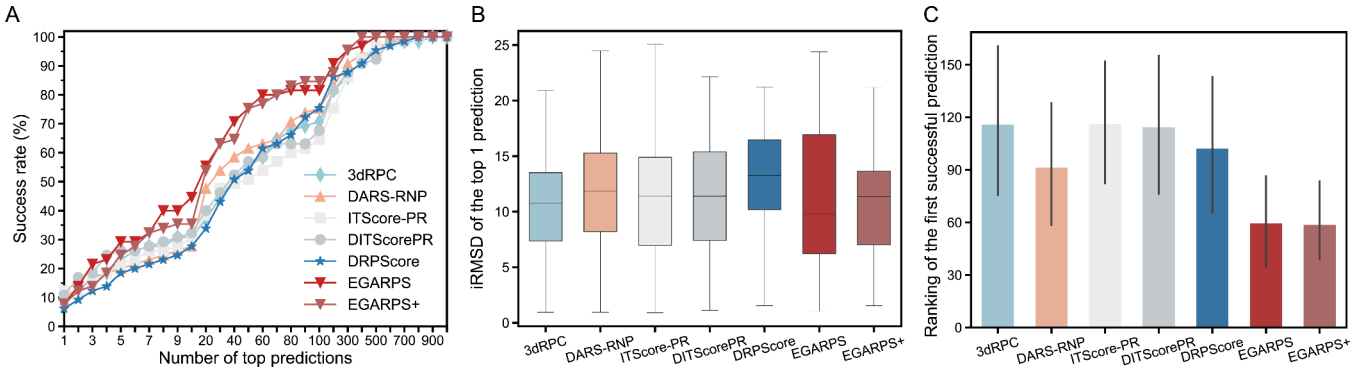


Figure 4. Performance comparison between EGARPS series and other methods using the "fold-then-dock" strategy. (A) Success rates. (B) Distribution of iRMSD for top 1 predictions. (C) Average ranking of the first successful predictions.

existing methods could be summarized from three aspects. First, our algorithm exhibits a higher tolerance to conformational changes. This could be evidenced by the performance on Perez-Cano et al.'s docking benchmark, which includes a total of 106 complexes.[37] For each complex, at least one binding partner is either in free form or modeled by homology. As shown in Figure 3B, the EGARPS series outperformed other methods in success rate, especially surpassing another deep learning model DRPScore by a large margin. The fine-grained methods, ITScore-PR and DITScorePR, performed less well than the coarse-grained method DARS-RNP, probably because the utility of atomic models is prone to be influenced by the local conformational changes involved in protein-RNA interactions. Second, EGARPS series is less sensitive to the interface size of native complexes. We divided all the samples into three groups according to the number of binding residues and nucleotides (small: 10∼40, medium: 40∼70, and large: 70∼120). Figure 3C shows the top1 success rates of various methods on the three groups. For testing sets I and II, EGARPS+ achieved the highest top 1 success rates on small interfaces (0.67 and 0.25, respectively), while the second-best method (DITScorePR) achieved the values of 0.52 and 0.19, respectively. Regarding testing set III, our two models achieved a top 1 success rate of 0.12 on small interfaces, whereas other methods failed to make successful predictions. Supplementary Figure 6 presents a complex with 26 binding residues and nucleotides. EGARPS+ achieved an iRMSD of 1.04 Å, while iRMSD values of other methods were close to or greater than 10 Å. Third, the proposed method exhibits better generalizability on complexes sharing less structure similarities with training samples. We classified the testing samples into three groups according to the TM-scores between each testing complex and all the training complexes (difficult: 0∼0.45, medium: 0.45∼0.75, and easy: 0.75∼1).[38] Figure 3D shows the successful number of top 1 predictions for the three deep learning-based methods. For easy samples, the EGARPS series achieved a successful rate of approximately 50%, compared to 33% for DRPScore. For medium samples, the performance of DRPScore was reduced, while EGARPS series obtained acceptable results on three testing sets. For difficult samples, our methods yielded a success rate of 50% on the testing set II, while DRPScore achieved a rate of 30%. Moreover, both EGARPS and EGARPS+ recovered two complexes from the other two data sets, whereas DRPScore lost its prediction ability on these partitions. To assess the generalizability of our models on unseen samples (i.e., zero-shot evaluation), we excluded the complexes in our three testing sets, in which the protein chains share the same Pfam domains with the training complexes. For the remaining samples, EGARPS series still showed better performance than existing algorithms (Supplementary Figure 7). Additionally, we collected a new data set of 9 unseen complexes (Supplementary Text 1). Our methods performed worse than atom-based statistical potential methods but outperformed residue-based statistical potential methods and DRPScore. Collectively, these advantages enable EGARPS series to be more adaptable to real scenarios.

Performance of the "Fold-Then-Dock" Strategy. Here, we also evaluated the "fold-then-dock" strategy using testing sets. First, we generated the predicted structures of protein and RNA in each complex by the trRosetta and trRosettaRNA programs, respectively.[39−42] Then, 1000 decoys were generated for each pair of predicted structures using 3dRPC. If the iRMSD values of all decoys were greater than 4 Å, this complex was excluded. As a result, 65 complexes were retained to assess the performance of scoring functions. Both our models yielded the top 1 success rates of 7.69%, which was slightly inferior to the measures of three statistical methods (e.g., ITScore-PR and DARS-RNP). However, EGARPS achieved the best top 3 success rate among all methods, and our models showed clearly better performance from the top 5 to top 500 predictions (Figure 4A). Regarding the iRMSD of top 1 prediction, only the median of EGARPS was less than 10 Å, while DRPScore obtained the greatest median of 13.23 Å (Figure 4B). Note that this measure of our models was clearly lower when using the native structure for docking (Supplementary Figure 8). According to the first successful predictions, EGARPS and EGARPS+ yielded the average rankings of 60 and 59, respectively, compared to 91 for DRPS-SNP and 116 for ITScore-PR (Figure 4C). Thus, all these scoring functions were effective in the "fold-then-dock" strategy, and the EGARPS series still maintained their advantages. In addition, we collected 13 RNA−protein complexes with a resolution between 3.5 and 5 Å. For the top 1 prediction, EGARPS was inferior to ITScore-PR but was superior to other methods (Supplementary Text 1 and Supplementary Figure 9). The results suggested that our scoring function could also be applicable to ambiguous complexes.

Potential to Improve De Novo Prediction Accuracy. Recently, RoseTTAFoldNA was developed for de novo structure prediction of protein-nucleic acid complexes.[43] However, its built-in assessment measure used to rank predicted structural models may not be the optimal way. Here, we adopted our methods to re-evaluate the quality of

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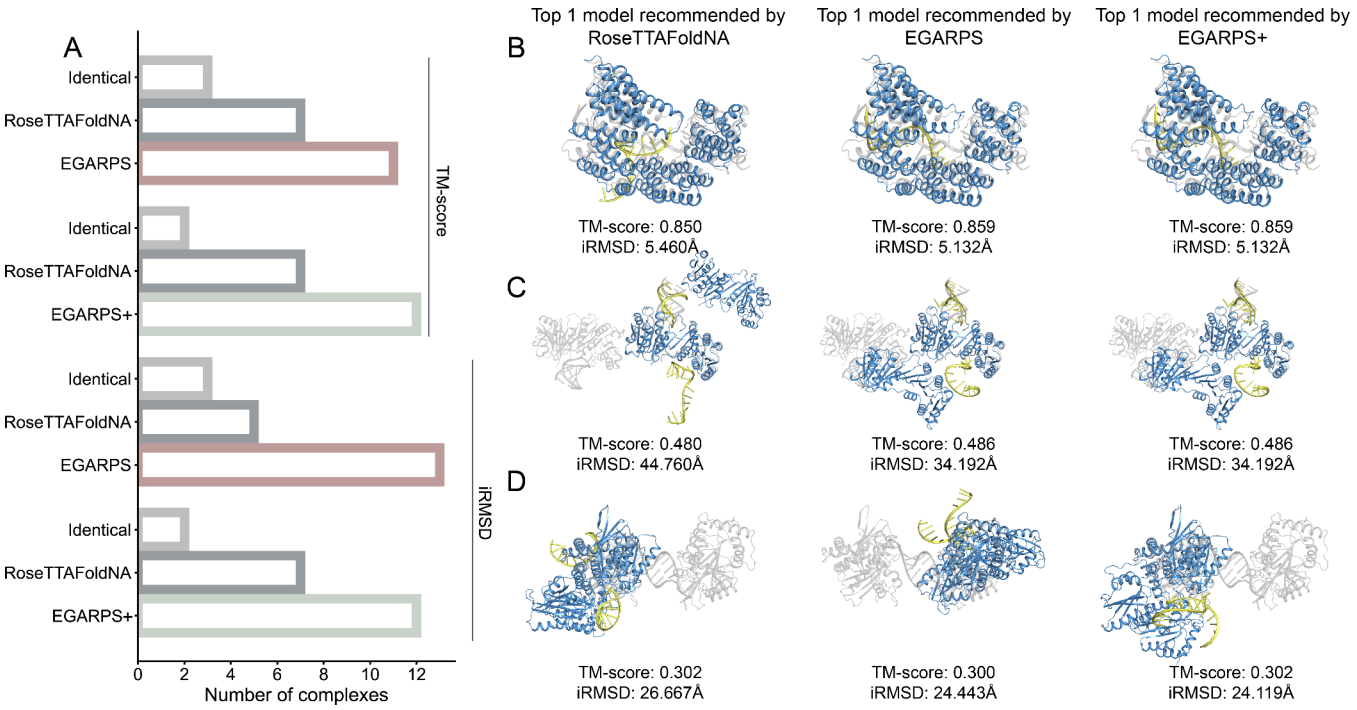


Figure 5. Comparison between our method and the built-in assessment method of RoseTTAFoldNA. (A) Numbers of top 1 predictions having better TM-score or iRMSD measures. (B–D) Top 1 predictions of three examples (PDB IDs: 7A9W, 8H0S, and 7UU3) generated by different methods. Native proteins and RNAs are colored in gray, while recommended proteins and RNAs are colored in blue and yellow, respectively.

structural models output by RoseTTAFoldNA. To this end, we collected RNA–protein complexes solved by X-ray crystallography after May first, 2020 from the PDB, which were not in the training set of RoseTTAFoldNA. After removing the redundancy relative to our training set, we retained 21 RNA–protein complexes (Supplementary Table 4). For each complex, ten candidate conformations were generated by RoseTTAFoldNA and then reranked by EGARPS and EGARPS+. Figure 5A shows the number of top-ranked conformations with a better TM-score or iRMSD. It is evident that EGARPS series outperformed the built-in assessment of RoseTTAFoldNA. As illustrated in Figure 5B–D, regardless of the prediction quality of structural models by RoseTTAFoldNA, the EGARPS series was able to distinguish conformations closer to the native structures. Additionally, we conducted a similar analysis using AlphaFold3[44] (Supplementary Figure 10). Compared to RoseTTAFoldNA, AlphaFold3 exhibited better prediction qualities, and its built-in assessment measure considered both local and global accuracies. In this scenario, the EGARPS series still held moderate advantages. Based on TM-score, the numbers of better conformations were 12 and 8 for EGARPS+ and AlphaFold3, respectively. In terms of iRMSD, EGARPS and AlphaFold3 retrieved 11 and 9 better conformations, respectively. Furthermore, we replaced our methods with existing scoring functions to rerank these conformations. As shown in Supplementary Figure 11, the EGARPS series exhibited superiority across nearly all cases, and competing methods rarely yielded better results for more than 10 complexes. Thus, our method could serve as a powerful reranking tool to assist in the de novo structure prediction of RNA–protein complexes.

Interpretability and Ablation Studies of Our Algorithm. We explored the interpretability of our model by analyzing how its attention values change in response to different decoys. We first compared the attention values from EGARPS+ for the binding interface of decoys with different sequential conservations.[45] For the molecular attention, as presented in Supplementary Figure 12A,B, attention values of sequence contexts were obviously higher for interfaces possessing lower entropy values (i.e., stronger conservation), while those of structural contexts were generally lower in this scenario. As the sequential conservation of binding regions became weaker, the attention for sequence contexts gradually declined and the role of structural contexts became relatively important. This result implies that our algorithm adaptively adjusts the contribution of sequence and structural information according to the presence and absence of conserved sequence patterns. Additionally, because hydrogen bonding (HB) is essential for RNA–protein interactions, we analyzed the attention values of HB-related nucleotides, residues, and nucleotide–residue pairs.[46] Supplementary Figure 12C,D shows that the nt-res pair pooling module assigned higher attention values to nucleotide-residue pairs forming hydrogen bonds. Similarly, higher values were given to nucleotides and residues serving as HB donors or acceptors by the nt/res attention pooling module. Thus, our method could identify key nucleotides and residues that support RNA–protein interactions.

Furthermore, the above tendency became more significant as iRMSD decreased, probably suggesting that our attention mechanisms were more effective in identifying critical contacts as the decoys were closer to native structures. Figure 6A displays a complex that contains the U1A-binding motif (i.e., "AUUGCAC") in RNA.[47] For the top-ranked decoy, EGARPS + assigned higher attention values to the sequence context as well as HB-related sites and pairs (Figure 6B). For the decoy with a ranking of 31, the interfacial segment still comprised the binding motif, thus maintaining relatively high attention values

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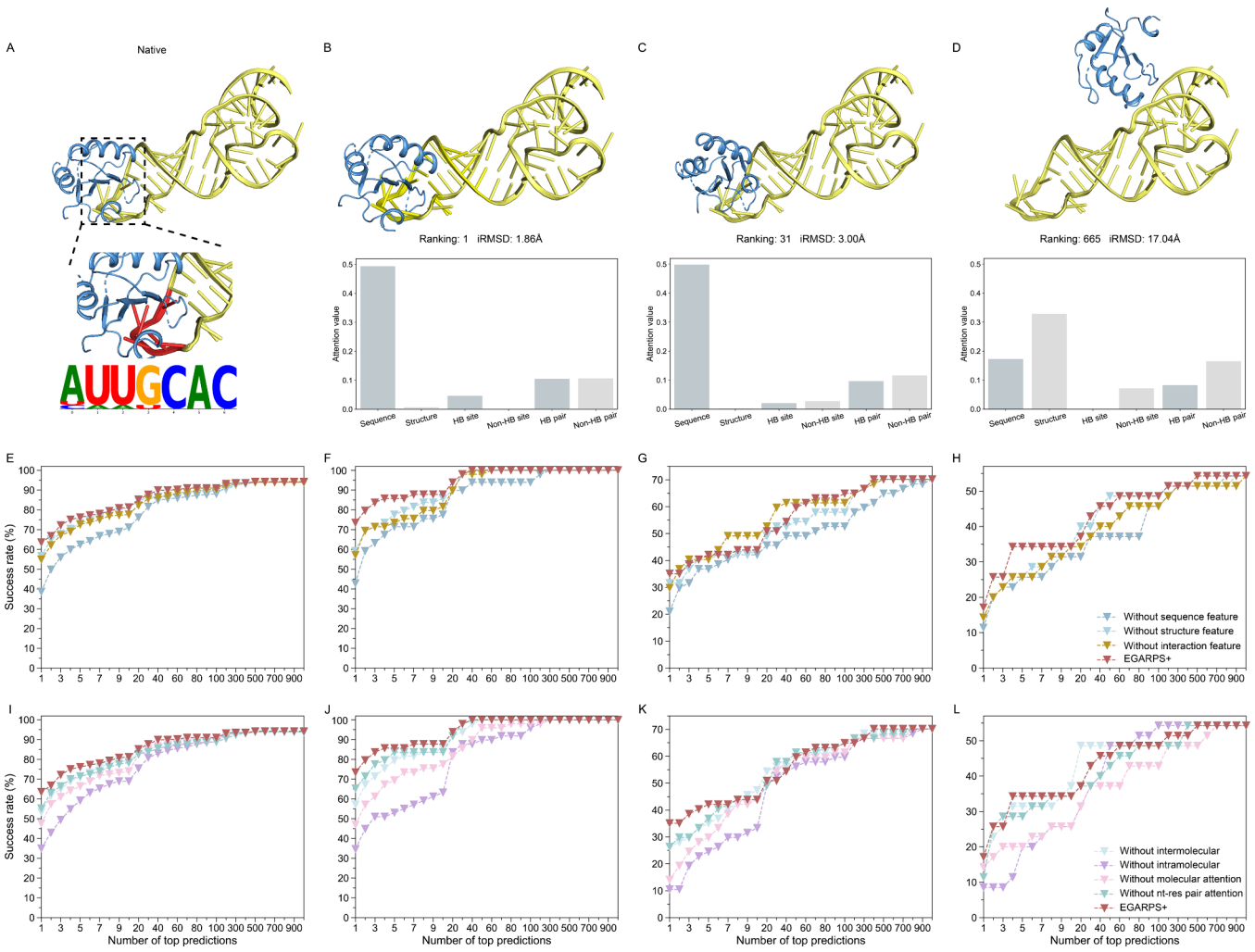


Figure 6. Interpretability of attention pooling modules and ablation studies of EGARPS+. (A) A native complex structure (PDB ID: 3EGZ) with a known binding motif. (B) Decoy with the first ranking and its attention values. (C) Decoy with the 31st ranking and its attention values. (D) Decoy with the 665th ranking and its attention values. Sequence/Structure denotes the attention between RNA and protein sequence/structure contexts, which is extracted from the molecular attention pooling. HB site denotes the attention to nucleotides/residues serving as hydrogen bonding donors or acceptors, which is extracted from the nt/res attention pooling. HB pair denotes the attention to nucleotide-residue pairs involving hydrogen bonds, which is extracted from the nt-res pair attention pooling. (E−H) Ablation studies for different types of features on training set and testing sets I, II and III, respectively. (I−L) Ablation studies for different modules on training set and testing sets I, II, and III, respectively.

for sequence contexts (Figure 6C). Although this structure was a correct prediction (iRMSD < 4 Å), lower attention values were assigned to HB-related sites by the nt/res and nt-res pair attention pooling. Therefore, our model could find changes in hydrogen-bonding patterns in partially plausible decoys, demonstrating its sensitivity to subtle structural discrepancies potentially affecting specific binding. For the decoy with a ranking of 665, in which the binding motif was absent, attention values for sequence contexts were markedly reduced, whereas more attention was given to structural contexts and non-HB sites and pairs (Figure 6D). Altogether, the above analyses illustrate the interpretability of our prediction framework.

By searching against the ATtRACT database, we retrieved 18 complexes from our testing sets, in which the native interface contains a known RBP-binding motif.[48] For the correct decoys, all residues were ranked by attention values. Supplementary Figure 13 displays that motif-interacting residues were significantly enriched among high-attention residues and had relatively greater attention values than other residues. Based on these complexes, we evaluated the three individual models with different seeds of EGARPS+. As shown in Supplementary Figure 14C, the success rates of these models were relatively stable. Although these models might select different decoys as top 1 predictions, they generally identified native-like structures having known motifs (Supplementary Table 5). Especially, EGARPS+ yielded a successful top 1 prediction for 10 complexes, among which 8 structures preserved the functional motif at the interface (Supplementary Figure 14D-K).

Furthermore, we conducted ablation studies at both the feature and module levels to assess the impact of different components on EGARPS+. As shown in Figure 6E−H, exclusion of sequence features induced the most significant decrease in performance, while structural and interaction features exhibited similar contributions. For instance, on the training set and testing set I, excluding sequence features resulted in a notable drop in the top 1 success rate from

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63.54% and 73.47% to 38.63% and 42.86%, respectively. This could be attributed to the importance of sequence conservation, which is a key factor correlated with molecular binding. As shown in Supplementary Figure 15, the embedding of language models may be the most important among the sequence features, and the atomic distance induced a marked influence on bound−bound structures among the interaction features. We also utilized the integrated gradients to perform the feature attribution analysis. These results were generally consistent with those of the ablation experiments (Supplementary Figure 16). Figure 6I−L suggest that the intramolecular module played a more crucial role than the intermolecular counterpart. When the two modules were separately removed, the top 1 success rates were 34.69% and 57.14% on testing set I and 10.53% and 26.32% on testing set II, respectively. Notably, the utility of the intramolecular module highly relied on the molecular attention pooling. After removing this operation, the top 1 success rate was reduced to 46.94% and 14.04% on testing sets I and II, respectively. By contrast, the influence of pair attention pooling was smaller, and its removal resulted in the values of 65.31% and 26.32%, respectively. Statistical differences between the ablation models and EGARPS+ are presented in Supplementary Text 2 and Supplementary Tables 6 and 7. Collectively, the results underscore the benefits of adopting a multifaceted perspective to explore RNA−protein binding conformations.

DISCUSSION

The application of deep learning to scoring RNA−protein complex structures is an emerging and promising field. Compared with traditional statistical potential-based methods, deep learning has shown advantages in handling conformational flexibility as demonstrated by DRPScore. This study proposes EGARPS+, an innovative graph learning-based scoring function in this field. Comprehensive evaluations indicated that our algorithm achieved the optimal overall performance and the highest top 1 success rate on various data sets. Relative to state-of-the-art methods, EGARPS+ could effectively handle the complexes involving greater conformational changes, smaller interface sizes and lower structural similarities to training samples. Besides, EGARPS+ had the ability to rerank the de novo predicted complex structures for improving the performance. The advantage of our approach is attributed to its GNN-based modular design. The development of intermolecular and intramolecular modules, coupled with the customization of suitable attention layers for these two modules, may be the crucial factor. The results showed that the intramolecular module played a major role in our framework, and the molecular attention pooling layer within this module had a greater impact than the nt-res pair attention pooling layer. The interpretability analysis suggested that the representation and learning of binding motifs and hydrogen bonding involved in RNA−protein interfaces is worthy of further exploration. Moreover, we analyzed the inference time of EGARPS+, which took approximately 90 s for a large complex of about 1000 nucleotides and residues, indicating that our algorithm could be applicable to large-scale computation (Supplementary Figures 17 and 18 and Supplementary Table 8). Despite the promising results, this prediction framework only focused on interacting regions and neglected noninteracting regions. In fact, noninteracting components also exert an effect on the formation of binding interfaces. In the future, we should consider the information on entire complexes, especially exploring the contribution of noninterface regions. In summary, we developed a novel scoring function called EGARPS+ that can effectively evaluate the structure of RNA−protein complexes. This tool might be helpful in uncovering the structural mechanism of RNA−protein interactions.

MATERIALS AND METHODS

Datasets. In this work, we adopted the training and testing sets prepared by Zeng et al. to evaluate our algorithm.[21] The RNA−protein complexes in the training set and the testing set I were extracted from the NDB database.[49] These samples satisfied the following criteria: the resolution of each complex was better than 3.5 Å, the length of an RNA was greater than or equal to 10 nucleotides, the number of chains in each entry was no more than 6, and the sequence identity of RNAs between and within the data sets was less than 95%.[50] The two data sets contained 277 and 49 bound−bound structures, respectively. Here, a bound−bound case means both RNA and protein structures were derived from the same complex. The complex structures in the testing set II were originally collected by Huang and Zou, including the unbound RNA and/or protein for docking and scoring.[51] The unbound structure means a binding partner either in free form or derived from a different complex. After removing complexes sharing high sequence similarity with the training set, this data set contained 41 unbound−unbound complexes and 16 unbound−bound complexes. To further assess the proposed algorithm, we constructed a more rigorous and challenging data set, namely the testing set III. The complexes in this data set were extracted from the docking benchmark built by Perez-Cano et al.[37] The unbound structure in this benchmark referred to a binding partner either in free form or modeled by homology. After reducing the sequence redundancy as mentioned above, we finally retained 35 complexes. A summary of the data sets is shown in Supplementary Table 9.

We used the 3dRPC package to generate RNA−protein decoys using the command "3dRPC-mode 9-system 8-par RPDOCK.par".[52] For each complex in the training set, the top 500 decoys out of 10,000 generated conformations were reserved based on the interface RMSD (iRMSD) ranking. The iRMSD measure was calculated through the command "3dRPC-mode 2-system 0-par RMSD.par". Regarding the test set, 1000 decoys were generated for each complex. The maximum, minimum, and average of iRMSD values of each complex are shown in Supplementary Table 10. For supervised learning, the native structures were used as positive samples, and the decoys were used as negative samples.

Proposed Algorithm. In this work, we extracted multifaceted features to represent an RNA−protein interface and then integrated EGNNs, GRUs, and attention pooling to develop prediction models. The features comprised the sequence and structural descriptors of nucleotides/residues and the interaction descriptors of pairwise nucleotides/residues. For the intermolecular graph, EGNNs were trained to optimize node features (i.e., sequence and structural embedding) and edge features (i.e., interaction embedding), and the nt/res attention was used to assess the importance of nodes. For the intramolecular graph, EGNNs and GRUs were used to update the nucleotide/residue embeddings. Here, the molecular attention checked the complementarity between sequence/structural contexts, while the pair attention evaluated the association between nucleotides and residues.

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Finally, the pooling results generated by intermolecular and intramolecular modules were combined to predict the probability score. We also explored alternative strategies to build the prediction models, but the results were inferior to the performance of the proposed architecture (Supplementary Text 3 and Supplementary Figure 19).

Feature Extraction. *Sequence Features. Nucleotide (Residue) Type Representation.* One-hot encoding is an effective method for representing biological sequences. In this study, we applied this encoding technique to the four types of nucleotides and 20 types of amino acids, thus generating one-hot representations of RNA and protein sequences.

Language Model Embedding. Currently, the large language models trained on massive biological sequences show the ability to produce representations that effectively capture structural and functional information. Among existing language models, we selected RNA−FM and ESM2 to generate the representations of RNA and protein sequences, respectively.[53,54]

Relative Position. The relative position of a nucleotide (residue) in an RNA (protein) sequence can be calculated using the following formula:[55]

$$y = \frac{x_n}{L}$$

where x_n is the position of the nth nucleotide (residue) in the sequence and L is the length of the sequence.

Structural Features. Secondary Structure. The DSSR and DSSP programs were used to assign the secondary structures of RNA and protein, respectively.[56,57] We considered seven states for secondary structures of RNAs, including nonloop single-stranded segment, stem, hairpin, bulge, internal loop, multiple loop, and pseudoknot. Meanwhile, we adopted three states for secondary structures of proteins, namely, helix, strand, and coil. Each nucleotide or residue was represented by a corresponding one-hot vector.

Backbone Geometry. To capture the geometric features of the backbone, we computed six pseudo torsion angles (i.e., η, θ, η', θ', η'', and θ'') for RNA nucleotides.[58,59] Moreover, we calculated two backbone torsion angles (i.e., Φ and ψ) for protein residues.[60] Then, the sine and cosine transformations were applied to these torsion angles.

Tertiary Structure. Based on our previous experiences, we used Laplacian norms (LNs) as global features and selected the relative solvent accessibility and geometry-based descriptors as local features to reflect the structural characteristics of each nucleotide or residue in a binding partner. The LN measure was defined as the distance between the target nucleotide (residue) and the weighted center of its surrounding context.[61] The formula is presented as follows:

$$\ln_i(\sigma) = \left\| c_i - \frac{\sum_j^{|i-j|>1} [c_j \times \Omega_{ij}(\sigma)]}{\sum_j^{|i-j|>1} \Omega_{ij}(\sigma)} \right\|$$

where c_i and c_j denote the average coordinates of all the atoms in nucleotides (residues) i and j. $\Omega_{ij}(\sigma)$ is the Laplace operator which can be calculated as follows:

$$\Omega_{ij}(\sigma) = \begin{cases} e^{-\|c_i - c_j\|^2/\sigma^2} & \text{if} \quad |i-j|>1 \\ 0 & \text{otherwise} \end{cases}$$

where σ is the scale factor and determines the weight of each nucleotide (residue) in a given context. Here, we chose the 0, 1/4, 1/2, 3/4, and 1 quantiles of the distance distribution of all the paired nucleotides (residues) as the scale factors. The relative solvent accessible surface area was calculated using the NACCESS program with a probe diameter of 1.5 Å.[62] The local geometric descriptors, including the minimum inaccessible radius, accessible shell volume, and pocketness, were computed by the GHECOM program.[63]

Interaction Features. Following the work of Morehead et al., we adopted rotation- and translation-invariant features (i.e., the distance, direction, and orientation) between a pair of residues/nucleotides as relative geometry.[64,65] Specifically, we built a local coordinate system N_i for the ith nucleotide as follows:

$$N_i = [b_i, n_i, b_i \times n_i]$$

$$u_i = v_{O_5'} - v_{C_3'}, \quad v_i = v_{C_5'} - v_{C_3'}, \quad b_i = \frac{u_i}{\|u_i\|}, \quad n_i = \frac{u_i \times v_i}{\|u_i \times v_i\|}$$

where $v_{O_5'}$, $v_{C_3'}$, and $v_{C_5'}$ are vectors pointing from the origin to the atom O_5', C_3', and C_5', respectively. For the ith residue, the local coordinate system r_i was constructed as follows:

$$R_i = [b_i, n_i, b_i \times n_i]$$

$$u_i = v_C - v_{C_\alpha}, \quad v_i = v_N - v_{C_\alpha}, \quad b_i = \frac{u_i}{\|u_i\|}, \quad n_i = \frac{u_i \times v_i}{\|u_i \times v_i\|}$$

where v_C, v_{C_α}, and v_N are vectors pointing from the origin to the atom C, C_α, and N, respectively. Then, the distance, direction, and orientation measures for different types of interacting pairs could be calculated as follows:

$$\begin{cases} \left(\left\| v_{C_\alpha j} - v_{C_\alpha i} \right\|, \ R_i^T \dfrac{v_{C_\alpha j} - v_{C_\alpha i}}{\left\| v_{C_\alpha j} - v_{C_\alpha i} \right\|}, \ R_i^T \right) R_j & \text{for residue−residue pair} \\ \left(\left\| v_{C_3' j} - v_{C_3' i} \right\|, \ N_i^T \dfrac{v_{C_3' j} - v_{C_3' i}}{\left\| v_{C_3' j} - v_{C_3' i} \right\|}, \ N_i^T \right) N_j & \text{for nucleotide−nucleotide pair} \\ \left(\left\| v_{C_3' i} - v_{C_\alpha i} \right\|, \ N_i^T \dfrac{v_{C_3' i} - v_{C_\alpha i}}{\left\| v_{C_3' i} - v_{C_\alpha i} \right\|}, \ N_i^T \right) R_i & \text{for nucleotide−residue pair} \end{cases}$$

Furthermore, atom-based distances proposed by ITScore-PR were used as an additional edge feature in our enhanced algorithm (i.e., EGARPS+). Based on the atomic type classification of Babel, we calculated the distances between 12 RNA-derived atoms and 20 protein-derived atoms.[66]

Model Construction. *Equivariant Graph Neural Network.* We used a spatial k-nearest neighbor algorithm ($k = 12$) to construct the graph $g = (v, e)$, where v denotes interfacial nucleotides and/or residues, and e denotes the edges between nodes and their nearest neighbors. Each node had a group of scalar and vector features. Scalar features were the sequence and structural descriptors mentioned previously, and vector features were coordinates of nucleotides or residues in three-

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dimensional space. Additionally, the interaction descriptors were used as edge features. The EGNN layer could then be presented by the following formulas:

$$m_{ij} = \varnothing_e(h_i^l, h_j^l, \left\|x_i^l - x_j^l\right\|^2, a_{ij})$$

$$x_i^{l+1} = x_i^l + \frac{1}{|N(i)|}\sum_{j\neq i}(x_i^l - x_j^l)\varnothing_x(m_{ij})$$

$$m_i = \sum_{j\neq i} m_{ij}$$

$$h_i^{l+1} = \varnothing_h(h_i^l, m_i)$$

where h_i and h_j are the scalar features of nodes i and j, x_i and x_j are the vector features of nodes i and j, a_{ij} is the edge feature, and $N(i)$ is the neighbors of node i. $\varnothing_e$, $\varnothing_x$, and $\varnothing_h$ are shallow neural networks.

Gated Recurrent Unit. The sequence including all binding sites could be expressed as $S = \{..., v_{i-2}, v_{i-1}, v_i, v_{i+1}, v_{i+2}, ...\}$, where v_i denotes the ith interfacial nucleotide (residue) in the target RNA (protein). Each nucleotide or residue possessed a corresponding feature vector x_i, which was the concatenation of sequence and structural descriptors. Using $\{..., x_{i-2}, x_{i-1}, x_i, x_{i+1}, x_{i+2}, ...\}$ as input, we implemented the GRU layer as follows:

$$r_i = \sigma(W_r x_i + U_r h_{i-1} + b_r)$$

$$z_i = \sigma(W_z x_i + U_z h_{i-1} + b_z)$$

$$o_i = \tanh(W_h x_i + U_h(r_i \odot h_{i-1}) + b_h)$$

$$h_i = z_i \odot h_{i-1} + (1 - z_i) \odot o_i$$

where r_i is the reset gate, z_i is the update gate, h_{i-1} is the previous hidden state, o_i is the new hidden state, and σ is the sigmoid function. $\odot$ represents the element-wise multiplication. W and U are weight matrices, and b is a bias.

Nt/res Attention Pooling. We used $\{x_1, x_2, ..., x_n\}$ to denote the feature vectors of nodes in the graph of interfacial nucleotides and residues. Then, the nt/res attention pooling mechanism could be conducted as follows:

$$r_{\text{nrap}} = \sum_{i=1}^{n} \text{softmax}(Wx_i + b)x_i$$

where W is a learnable parameter matrix and b is a bias. This operation could reflect the unequal contributions of nodes in the graph, thus emphasizing the nucleotides and residues that played pivotal roles in the interactions.

Molecular Attention Pooling. Inspired by the work of Behera et al. on image processing, we developed a molecular attention pooling approach to capture the complementarity and matching between the binding regions of RNA and protein.[67] To this end, we constructed a context vector set, $C = \{p_{\text{str}}, p_{\text{seq}}, r_{\text{str}}, r_{\text{seq}}\}$, which contains the structural and sequence context vectors of binding regions. The structural context was the mean aggregation of node vectors updated through EGNN, and the sequence context was the mean aggregation of the sequential outputs from GRU. Each context vector could be transformed into the weighted sum of itself and the remaining vectors using the following formulas:

$$c_r = \sum_{r'\in C} \alpha_{r,r'} r'$$

$$\alpha_{r,r'} = \text{softmax}(W_\alpha \beta_{r,r'} + b_\alpha)$$

$$\beta_{r,r'} = \tanh(q_r + k_{r'} + b_\beta)$$

$$q_r = W_q r \quad \text{and} \quad k_{r'} = W_k r'$$

where W and b are learnable parameters, $r \in C$ and $r' \in C$. Then, a learnable pooling operation was performed as follows:

$$r_{\text{map}} = \sum_{r\in C} \text{softmax}(Wc_r + b)c_r$$

Nt-res Pair Attention Pooling. In contrast to the molecular attention pooling, the nt-res pair attention pooling focused on nucleotides and residues that have a high probability to interact with each other.[68] Let $R = \{r_1, r_2, ..., r_n\}$ denote feature vectors of interfacial nucleotides in an RNA and $P = \{p_1, p_2, ..., p_m\}$ denote feature vectors of interfacial residues in a protein. Conditioned on P, the vectors of interfacial nucleotides were updated in two steps. First, the attention scores between interfacial nucleotides and residues were calculated as follows:

$$a_{ij} = ReLU(b_i^T b_j)$$

$$b_i = Wf_i \quad \text{and} \quad b_j = Wf_j$$

where W is a learnable parameter matrix, $f_i \in R$ and $f_j \in P$. The dot-product attention score might capture the association between hidden representations. Then, the updated vectors of interfacial nucleotides were calculated using the vectors of interfacial residues along with the normalized attention scores as follows:

$$c_i = \sum_{j=1}^{m} \alpha_{ij} f_j$$

$$\alpha_{ij} = \frac{a_{ij}}{\sqrt{\sum_{i,j=1}^{i=n,j=m} a_{ij}^2}}$$

Similarly, we could update the vectors of interfacial residues conditioned on R using the above procedure. Then, the pooling representation was calculated as follows:

$$r_{pap} = \frac{\sum_{i=1}^{i=n} c_i + \sum_{j=1}^{j=m} c_j}{|R| + |P|}$$

Multilayer Perceptron. For each interface, three types of embedding (i.e., the outputs of three different attention pooling layers) were concatenated and fed into the multilayer perceptron to obtain the probability of being a native-like complex:

$$y = \text{Sigmoid}(W_3(W_2(\text{ReLU}(W_1[r_{\text{nrap}}\|r_{\text{map}}\|r_{\text{pap}}] + b_1)) + b_2) + b_3)$$

where W_1, W_2, and W_3 are weight matrices, and b_1, b_2, and b_3 are biases. The ReLU function was applied to the first layer to implement a nonlinear transformation. The sigmoid function was used to convert the prediction values to a range of 0 to 1.

Implementation Details. We performed 5-fold cross-validation on the training set to determine the optimal hyperparameters of our method. During the training process, each batch contained one native structure and one decoy structure. We employed the RAdam optimizer with a learning

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rate of 0.00001 for model optimization guided by the binary cross entropy loss.[69] The training procedure lasted for 100 epochs on the GeForce RTX 3080. This process was repeated once and three times for EGARPS and EGARPS+, respectively. The PyTorch and deep graph library were used to implement the deep learning framework.[70,71]

Evaluation Metrics. We adopted the iRMSD measure suggested by the CAPRI competition to evaluate the quality of predicted complex structures.[72] This measure reflects the difference in interface regions between the predicted and native structures. We utilized the 3dRPC program to superimpose two complexes and calculate the iRMSD measure. An RNA−protein complex was considered successfully predicted if the iRMSD between the predicted and native structures was no more than 4 Å. The top N success rate was used to evaluate our algorithm and make comparisons with existing methods.

■ ASSOCIATED CONTENT

Data Availability Statement

All data sets used in this study can be downloaded from the following link: https://github.com/hzau-liulab/EGARPS.

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.jctc.5c00831.

Additional experimental details, materials, and evaluations, including data set summaries, model architecture, comparison with existing methods, statistical test, analysis of attention values, ablation analysis, and runtime comparisons (PDF)

Accession Codes

All source codes and models of EGARPS are available at the following link: https://github.com/hzau-liulab/EGARPS.

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#These authors contributed equally to this work (Z.J. and Y.Z.).

Notes

The authors declare no competing financial interest.

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■ REFERENCES

(1) Lunde, B. M.; Moore, C.; Varani, G. RNA-binding proteins: modular design for efficient function. *Nat. Rev. Mol. Cell Biol.* **2007**, *8* (6), 479−490.

(2) Licatalosi, D. D.; Darnell, R. B. RNA processing and its regulation: global insights into biological networks. *Nat. Rev. Genet* **2010**, *11* (1), 75−87.

(3) Müller-McNicoll, M.; Neugebauer, K. M. How cells get the message: dynamic assembly and function of mRNA-protein complexes. *Nat. Rev. Genet* **2013**, *14* (4), 275−287.

(4) He, P. C.; He, C. mRNA accessibility within mRNPs as a determinant of gene expression. *Trends Biochem. Sci.* **2024**, *49* (3), 199−207.

(5) Modic, M.; Ule, J.; Sibley, C. R. CLIPing the brain: studies of protein-RNA interactions important for neurodegenerative disorders. *Mol. Cell Neurosci* **2013**, *56*, 429−435.

(6) Gebauer, F.; Schwarzl, T.; Valcárcel, J.; Hentze, M. W. RNA-binding proteins in human genetic disease. *Nat. Rev. Genet* **2021**, *22* (3), 185−198.

(7) Ruffenach, G.; Medzikovic, L.; Sun, W.; Hong, J.; Eghbali, M. Functions of RNA-binding proteins in cardiovascular disease. *Cells* **2023**, *12* (24), 2794.

(8) Zhao, H.; Yang, Y.; Zhou, Y. Highly accurate and high-resolution function prediction of RNA binding proteins by fold recognition and binding affinity prediction. *RNA Biol.* **2011**, *8* (6), 988−996.

(9) Obayashi, E.; Oubridge, C.; Krummel, D. P.; Nagai, K. Crystallization of RNA-protein complexes. *Methods Mol. Biol.* **2007**, *363*, 259−276.

(10) Scott, L. G.; Hennig, M. RNA structure determination by NMR. *Methods Mol. Biol.* **2008**, *452*, 29−61.

(11) Henderson, R. From Electron Crystallography to Single Particle CryoEM (Nobel Lecture). *Angew. Chem., Int. Ed. Engl.* **2018**, *57* (34), 10804−10825.

(12) Laing, C.; Schlick, T. Computational approaches to 3D modeling of RNA. *J. Phys.: Condens. Matter* **2010**, *22* (28), 283101.

(13) Rother, K.; Rother, M.; Boniecki, M.; Puton, T.; Bujnicki, J. M. RNA and protein 3D structure modeling: similarities and differences. *J. Mol. Model* **2011**, *17* (9), 2325−2336.

(14) Tuszynska, I.; Matelska, D.; Magnus, M.; Chojnowski, G.; Kasprzak, J. M.; Kozlowski, L. P.; Dunin-Horkawicz, S.; Bujnicki, J. M. Computational modeling of protein-RNA complex structures. *Methods* **2014**, *65* (3), 310−319.

(15) Qiu, L.; Zou, X. Scoring functions for protein-RNA complex structure prediction: advances, applications, and future directions. *Commun. Inf Syst* **2020**, *20* (1), 1−22.

(16) Pérez-Cano, L.; Solernou, A.; Pons, C.; Fernández-Recio, J. Structural prediction of protein-RNA interaction by computational docking with propensity-based statistical potentials. *Pac Symp. Biocomput* **2009**, 293−301.

(17) Tuszynska, I.; Bujnicki, J. M. DARS-RNP and QUASI-RNP: new statistical potentials for protein-RNA docking. *BMC Bioinformatics* **2011**, *12*, 348.

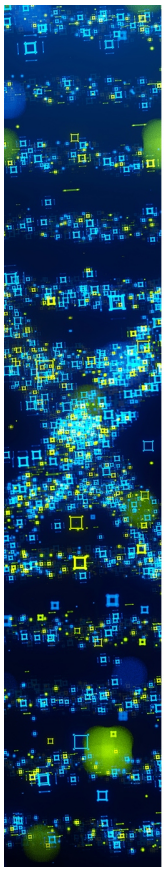
(18) Li, H.; Huang, Y.; Xiao, Y. A pair-conformation-dependent scoring function for evaluating 3D RNA-protein complex structures. *PLoS One* **2017**, *12* (3), No. e0174662.

(19) Huang, S. Y.; Zou, X. A knowledge-based scoring function for protein-RNA interactions derived from a statistical mechanics-based iterative method. *Nucleic Acids Res.* **2014**, *42* (7), e55.

(20) Parisien, M.; Wang, X.; Perdrizet, G., 2nd; Lamphear, C.; Fierke, C. A.; Maheshwari, K. C.; Wilde, M. J.; Sosnick, T. R.; Pan, T. Discovering RNA-protein interactome by using chemical context

profiling of the RNA-protein interface. *Cell Rep* **2013**, *3* (5), 1703−1713.
(21) Zeng, C.; Jian, Y.; Vosoughi, S.; Zeng, C.; Zhao, Y. Evaluating native-like structures of RNA-protein complexes through the deep learning method. *Nat. Commun.* **2023**, *14* (1), 1060.
(22) Dauparas, J.; Anishchenko, I.; Bennett, N.; Bai, H.; Ragotte, R. J.; Milles, L. F.; Wicky, B. I. M.; Courbet, A.; de Haas, R. J.; Bethel, N.; et al. Robust deep learning-based protein sequence design using ProteinMPNN. *Science* **2022**, *378* (6615), 49−56.
(23) Xia, Y.; Xia, C. Q.; Pan, X.; Shen, H. B. GraphBind: protein structural context embedded rules learned by hierarchical graph neural networks for recognizing nucleic-acid-binding residues. *Nucleic Acids Res.* **2021**, *49* (9), e51.
(24) Gligorijević, V.; Renfrew, P. D.; Kosciolek, T.; Leman, J. K.; Berenberg, D.; Vatanen, T.; Chandler, C.; Taylor, B. C.; Fisk, I. M.; Vlamakis, H.; et al. Structure-based protein function prediction using graph convolutional networks. *Nat. Commun.* **2021**, *12* (1), 3168.
(25) Lu, C.; Lubin, J. H.; Sarma, V. V.; Stentz, S. Z.; Wang, G.; Wang, S.; Khare, S. D. Prediction and design of protease enzyme specificity using a structure-aware graph convolutional network. *Proc. Natl. Acad. Sci. U. S. A.* **2023**, *120* (39), No. e2303590120.
(26) Zhang, W.; Cui, Y.; Liu, B.; Loza, M.; Park, S. J.; Nakai, K. HyGAnno: hybrid graph neural network-based cell type annotation for single-cell ATAC sequencing data. *Brief Bioinform.* **2024**, *25* (3), bbae152.
(27) Franke, J. K. H.; Runge, F.; Hutter, F. Scalable Deep Learning for RNA Secondary Structure Prediction. *arXiv:2307.10073 [cs.LG]* **2023**, na.
(28) Ganea, O.-E.; Huang, X.; Bunne, C.; Bian, Y.; Barzilay, R.; Jaakkola, T.; Krause, A. Independent SE(3)-Equivariant Models for End-to-End Rigid Protein Docking. *arXiv:2111.07786 [cs.AI]* **2021**, na.
(29) Liu, R.; Hu, J. DNABind: a hybrid algorithm for structure-based prediction of DNA-binding residues by combining machine learning- and template-based approaches. *Proteins* **2013**, *81* (11), 1885−1899.
(30) Yang, X. X.; Deng, Z. L.; Liu, R. RBRDetector: improved prediction of binding residues on RNA-binding protein structures using complementary feature- and template-based strategies. *Proteins* **2014**, *82* (10), 2455−2471.
(31) Jiang, Z.; Shen, Y. Y.; Liu, R. Structure-based prediction of nucleic acid binding residues by merging deep learning- and template-based approaches. *PLoS Comput. Biol.* **2023**, *19* (9), No. e1011428.
(32) Jiang, Z.; Xiao, S. R.; Liu, R. Dissecting and predicting different types of binding sites in nucleic acids based on structural information. *Brief Bioinform* **2022**, *23* (1), bbab411.
(33) Li, Y.; Tarlow, D.; Brockschmidt, M.; Zemel, R. Gated graph sequence neural networks. *arXiv:1511.05493 [cs.LG]* **2015**, na.
(34) Garcia Satorras, V.; Hoogeboom, E.; Welling, M. E(n) equivariant graph neural networks. *arXiv:2102.09844 [cs.LG]* **2021**, na.
(35) Chung, J.; Gulcehre, C.; Cho, K.; Bengio, Y. Empirical evaluation of gated recurrent neural networks on sequence modeling. *arXiv:1412.3555 [cs.NE]* **2014**, na.
(36) He, J.; Tao, H.; Huang, S. Y. Protein-ensemble-RNA docking by efficient consideration of protein flexibility through homology models. *Bioinformatics* **2019**, *35* (23), 4994−5002.
(37) Pérez-Cano, L.; Jiménez-García, B.; Fernández-Recio, J. A protein-RNA docking benchmark (II): extended set from experimental and homology modeling data. *Proteins* **2012**, *80* (7), 1872−1882.
(38) Zhang, C.; Shine, M.; Pyle, A. M.; Zhang, Y. US-align: universal structure alignments of proteins, nucleic acids, and macromolecular complexes. *Nat. Methods* **2022**, *19* (9), 1109−1115.
(39) Du, Z.; Su, H.; Wang, W.; Ye, L.; Wei, H.; Peng, Z.; Anishchenko, I.; Baker, D.; Yang, J. The trRosetta server for fast and accurate protein structure prediction. *Nat. Protoc* **2021**, *16* (12), 5634−5651.
(40) Wang, W.; Peng, Z.; Yang, J. Single-sequence protein structure prediction using supervised transformer protein language models. *Nat. Comput. Sci.* **2022**, *2* (12), 804−814.
(41) Su, H.; Wang, W.; Du, Z.; Peng, Z.; Gao, S. H.; Cheng, M. M.; Yang, J. Improved Protein Structure Prediction Using a New Multi-Scale Network and Homologous Templates. *Adv. Sci. (Weinh)* **2021**, *8* (24), No. e2102592.
(42) Wang, W.; Feng, C.; Han, R.; Wang, Z.; Ye, L.; Du, Z.; Wei, H.; Zhang, F.; Peng, Z.; Yang, J. trRosettaRNA: automated prediction of RNA 3D structure with transformer network. *Nat. Commun.* **2023**, *14* (1), 7266.
(43) Baek, M.; McHugh, R.; Anishchenko, I.; Jiang, H.; Baker, D.; DiMaio, F. Accurate prediction of protein-nucleic acid complexes using RoseTTAFoldNA. *Nat. Methods* **2024**, *21* (1), 117−121.
(44) Abramson, J.; Adler, J.; Dunger, J.; Evans, R.; Green, T.; Pritzel, A.; Ronneberger, O.; Willmore, L.; Ballard, A. J.; Bambrick, J.; et al. Accurate structure prediction of biomolecular interactions with AlphaFold 3. *Nature* **2024**, *630* (8016), 493−500.
(45) Cook, K. B.; Kazan, H.; Zuberi, K.; Morris, Q.; Hughes, T. R. RBPDB: a database of RNA-binding specificities. *Nucleic Acids Res.* **2011**, *39*, D301−308.
(46) McDonald, I. K.; Thornton, J. M. Satisfying hydrogen bonding potential in proteins. *J. Mol. Biol.* **1994**, *238* (5), 777−793.
(47) Park, B.; Han, K. Discovering protein-binding RNA motifs with a generative model of RNA sequences. *Comput. Biol. Chem.* **2020**, *84*, 107171.
(48) Giudice, G.; Sánchez-Cabo, F.; Torroja, C.; Lara-Pezzi, E. ATtRACT-a database of RNA-binding proteins and associated motifs. *Database (Oxford)* **2016**, *2016*, baw035.
(49) Coimbatore Narayanan, B.; Westbrook, J.; Ghosh, S.; Petrov, A. I.; Sweeney, B.; Zirbel, C. L.; Leontis, N. B.; Berman, H. M. The Nucleic Acid Database: new features and capabilities. *Nucleic Acids Res.* **2014**, *42*, D114−122.
(50) Li, W.; Godzik, A. Cd-hit: a fast program for clustering and comparing large sets of protein or nucleotide sequences. *Bioinformatics* **2006**, *22* (13), 1658−1659.
(51) Huang, S. Y.; Zou, X. A nonredundant structure dataset for benchmarking protein-RNA computational docking. *J. Comput. Chem.* **2013**, *34* (4), 311−318.
(52) Huang, Y.; Liu, S.; Guo, D.; Li, L.; Xiao, Y. A novel protocol for three-dimensional structure prediction of RNA-protein complexes. *Sci. Rep* **2013**, *3*, 1887.
(53) Lin, Z.; Akin, H.; Rao, R.; Hie, B.; Zhu, Z.; Lu, W.; Smetanin, N.; Verkuil, R.; Kabeli, O.; Shmueli, Y.; et al. Evolutionary-scale prediction of atomic-level protein structure with a language model. *Science* **2023**, *379* (6637), 1123−1130.
(54) Chen, J.; Hu, Z.; Sun, S.; Tan, Q.; Wang, Y.; Yu, Q.; Zong, L.; Hong, L.; Xiao, J.; Shen, T. Interpretable RNA foundation model from unannotated data for highly accurate RNA structure and function predictions. *arXiv:2204.00300 [q-bio.QM]* **2022**, na.
(55) Shuvo, M. H.; Karim, M.; Roche, R.; Bhattacharya, D. PIQLE: protein-protein interface quality estimation by deep graph learning of multimeric interaction geometries. *Bioinform Adv.* **2023**, *3* (1), vbad070.
(56) Kabsch, W.; Sander, C. Dictionary of protein secondary structure: pattern recognition of hydrogen-bonded and geometrical features. *Biopolymers* **1983**, *22* (12), 2577−2637.
(57) Lu, X. J.; Bussemaker, H. J.; Olson, W. K. DSSR: an integrated software tool for dissecting the spatial structure of RNA. *Nucleic Acids Res.* **2015**, *43* (21), e142.
(58) Olson, W. K. Configurational statistics of polynucleotide chains. An updated virtual bond model to treat effects of base stacking. *Macromolecules* **1980**, *13*, 721−728.
(59) Keating, K. S.; Humphris, E. L.; Pyle, A. M. A new way to see RNA. *Q. Rev. Biophys.* **2011**, *44* (4), 433−466.
(60) Ginn, H. M. Torsion angles to map and visualize the conformational space of a protein. *Protein Sci.* **2023**, *32* (4), e4608.
(61) Liu, H. F.; Liu, R. Structure-based prediction of post-translational modification cross-talk within proteins using comple-

mentary residue- and residue pair-based features. *Brief Bioinform* **2020**, *21* (2), 609−620.
(62) Hubbard, S. J.; Thornton, J. M. *NACCESS*, Computer Program; Department of Biochemistry and Molecular Biology, University College: London, 1993.
(63) Kawabata, T. Detection of multiscale pockets on protein surfaces using mathematical morphology. *Proteins* **2010**, *78* (5), 1195−1211.
(64) Morehead, A.; Chen, X.; Wu, T.; Liu, J.; Cheng, J. EGR: equivariant graph refinement and assessment of 3D protein complex structures. *arXiv:2205.10390 [cs.LG]* **2022**, na.
(65) Ingraham, J.; Garg, V. K.; Barzilay, R.; Jaakkola, T. Generative models for graph-based protein design. *Proceedings of the 33rd International Conference on Neural Information Processing Systems*; Curran Associates Inc., 2019; p 1417.
(66) O'Boyle, N. M.; Banck, M.; James, C. A.; Morley, C.; Vandermeersch, T.; Hutchison, G. R. Open Babel: An open chemical toolbox. *J. Cheminform* **2011**, *3*, 33.
(67) Behera, A.; Wharton, Z.; Hewage, P.; Bera, A. Context-aware attentional pooling (CAP) for fine-grained visual classification. *AAAI* **2021**, *35*, 929.
(68) Pittala, S.; Bailey-Kellogg, C. Learning context-aware structural representations to predict antigen and antibody binding interfaces. *Bioinformatics* **2020**, *36* (13), 3996−4003.
(69) Liu, L.; Jiang, H.; He, P.; Chen, W.; Liu, X.; Gao, J.; Han, J. On the variance of the adaptive learning rate and beyond. *arXiv:1908.03265 [cs.LG]* **2019**, na.
(70) Adam Paszke, S. G.; Massa, F.; Lerer, A.; Bradbury, J.; Chanan, G.; et al. Pytorch: an imperative style, high-performance deep learning library. *Adv. Neural Inf Process Syst* **2019**, *32*, 8026−8037.
(71) Wang, M.; Zheng, D.; Ye, Z.; Gan, Q.; Li, M.; Song, X.; Zhou, J.; Ma, C.; Yu, L.; Gai, Y. Deep Graph Library: a graph-centric, highly-performant package for graph neural networks. *arXiv:1909.01315 [cs.LG]* **2019**, na.
(72) Janin, J.; Henrick, K.; Moult, J.; Eyck, L. T.; Sternberg, M. J.; Vajda, S.; Vakser, I.; Wodak, S. J. CAPRI: a Critical Assessment of PRedicted Interactions. *Proteins* **2003**, *52* (1), 2−9.


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